

Ushtrime ND3f

1.1.49 Nën ç'kënd duhet të hedhet predha me kusht që lartësia maksimale e saj të jetë e barabartë me rrezen e hedhjes ($h=r$)?

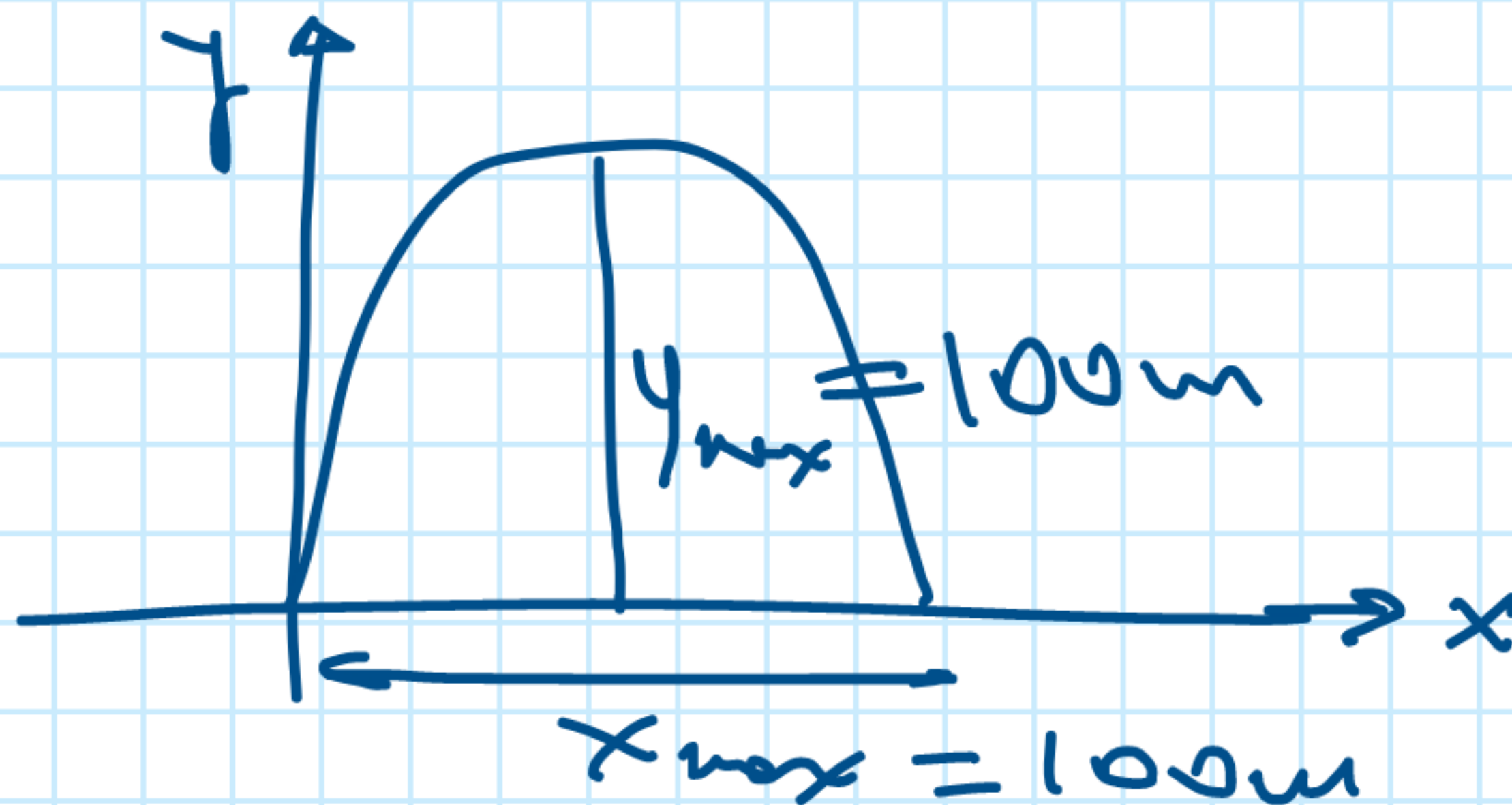
Zgjidhja:

Të dhënat:

$$h=r$$

$$y_{\max} = x_{\max}$$

$$\varphi = 75.9^\circ$$



$$\frac{v_0^2 \sin^2 \varphi}{2g} = \frac{v_0^2 \sin(2\varphi)}{g}$$

$$\frac{\sin^2 \varphi}{2} = 2 \sin \varphi \cos \varphi$$

$$\frac{\sin \varphi}{2} = 2 \cos \varphi \cdot \frac{2}{\omega \varphi}$$

$$\frac{\sin \varphi}{\cos \varphi} = 4$$

$$\tan \varphi = 4$$

$$\varphi = \arctan 4$$

$$\varphi = 75.9^\circ$$

$$x_{\max} = \frac{v_0^2 \sin 2\varphi}{g}$$

$$y_{\max} = \frac{v_0^2 \sin^2 \varphi}{2g}$$